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A Game Theory-Based Approach for Vehicle Platoon Merging at the Roundabout Intersection

Zhentao Zhang, Xiaoyin Zheng, Chuqi Su, Xun Liu, and Yiping Wang

***Abstract*—Connected and automated vehicles (CAVs) and platooning technologies represent promising solutions for mitigating urban traffic congestion. However, roundabout merging remains a significant challenge for CAVs. CAVs must negotiate with human drivers of heterogeneous styles to balance safety and efficiency, while simultaneously accounting for the impact of road curvature on vehicle dynamics and platoon stability. Existing research often oversimplifies this by modeling vehicles as point masses, ignoring the effects of driver reasoning, vehicle dynamics, and road geometry. To address these gaps, this paper proposes a game-theoretic platoon merging framework that integrates level-k reasoning with the principle of social optimality. All roundabout vehicles are mapped into virtual platoons via a coordinate transformation, and an artificial potential field that incorporates driving styles is devised to quantify interaction risks and guide merging with attractive forces. The framework dynamically assigns leader–follower roles based on predicted payoffs and social optimality to accelerate integration. Furthermore, the reasoning levels of surrounding vehicles are inferred using a Bayesian approach, allowing CAVs to adaptively adjust their own reasoning levels to prevent decision-making deadlocks. Simulation results demonstrate that the proposed method ensures safe and stable platoon operations at roundabout intersections and outperforms both the first-in-first-out virtual platoon strategy (FIFO-VPS) and the mixed-integer nonlinear programming (MINLP) approach in terms of merging efficiency and environmental adaptability.**

***Index Terms*—Autonomous driving, vehicle merging, game theory, potential field, roundabout intersection, Frenet coordinates.**

I. Introduction

IN RECENT years, an increasing number of policies have been enacted to support the testing and demonstration of connected and autonomous vehicles (CAVs) on public roads [1], [2]. However, complex road structures such as intersections, highway ramps and roundabouts pose significant challenges to the operation of CAVs [3], [4], [5]. In particular, while unsignalized roundabouts offer potential efficiency benefits, the sharp changes in road curvature and the increased number of conflict points necessitate a more nuanced consideration of control and planning requirements for CAVs [6], [7].

In urban traffic systems, the performance of the transportation network is largely dependent upon controlling areas prone to vehicle conflicts [8]. At roundabouts, conflicts typically arise when entering vehicles encounter vehicles already circulating within the roundabout [9]. Due to safety considerations, the priority for CAVs is generally lower than that for human drivers, which leads to a more conservative driving approach [10]. CAVs frequently encounter scenarios characterized by long waiting times or reciprocal yielding maneuvers with vehicles at roundabout intersections [11], [12]. The start-and-stop motion at these junctions not only contributes to traffic congestion but also increases the risk of accidents, thereby posing a safety concerns [13].

Currently, a substantial body of scholarly research has been dedicated to the merging issues of CAVs at intersections. These methodologies typically employ a hierarchical structure, where the upper layer module is responsible for determining the merging sequence of vehicles, and the lower layer module generates planned trajectories and control commands [14], [15]. A typical approach for addressing the merging problem for different lanes involves the concept of virtual platooning [16], [17]. By mapping vehicles from the mainline and ramp onto a single lane, the two-dimensional merging problem is transformed into a one-dimensional following problem [18]. However, this approach, similar to the first-in-first-out principle (FIFO), is intuitive but not necessarily optimal from a systemic perspective [19].

Vehicle merging represents a dynamic system, and researchers have reformulated it as an optimal control problem (OCP) to solve for the vehicle merging sequence. For example, Haoji Liu et al. determined the desired merging location of ramp CAVs by solving the unconstrained OCP using Pontryagin's minimum principle [20]. Shang-Chien Lin et al. used dynamic programming (DP) to determine the optimal passing order for a two-lane merging scenario [21]. Jieming Chen et al. employed a mixed-integer nonlinear programming (MINLP) approach to determine the optimal traffic sequence for autonomous vehicles at highway merging sections [22]. In addition to these methods,

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reinforcement learning (RL) has garnered widespread research interest within the academic community as an effective means to solve the vehicle merging problem [23], [24]. Compared to rule-based approaches, RL enables interactive agents to exhibit more human-like responsive behaviors. However, the challenge of weak interpretability and generalizability remains a significant hurdle in RL applications.

Due to the interactive game-playing between agents during the vehicle merging process, game theory is also widely utilized for modeling strategic behaviors in such scenarios [25], [26], [27]. For example, Li et al. proposed a novel dynamic game-theoretic model that transformed vehicle interactions at unsignalized intersections into multiple concurrent leader-follower subgame problems, thereby reducing computational complexity [28]. Wenli Li et al. employed a combination of game theory and inverse reinforcement learning techniques to predict the intersection behaviors of obstacle vehicles at intersections [29]. Liu et al. utilized noncooperative game theory to model collaborative vehicle platooning systems under both complete and incomplete information conditions [30]. In the context of game-theoretic interactions, the behavior of one agent has the potential to significantly influences the decision-making process of another. This mutual reasoning enables the prediction of other vehicles' responses to the ego vehicle's planned trajectory, allowing CAVs to operate more confidently and naturally. A typical model in this context is the level-k game, which treats the participants in the game as k-level reasoners, iteratively refining their level of reasoning [31].

Although extensive research has examined the safety and efficiency of vehicle merging, most studies focus on highway ramps or two-lane roads, while investigations of complex curved infrastructures such as roundabouts remain limited. Road geometry strongly influences trajectory prediction as well as lateral and longitudinal control. Simplifying vehicles as point masses to generate ideal trajectories overlooks vehicle dynamics and geometric constraints, making such trajectories impractical in real-world environments. In addition, the heterogeneous behaviors of surrounding drivers—shaped by diverse driving styles and reasoning abilities—further complicate CAV decision-making. To operate effectively, CAVs must anticipate the social consequences of their actions and respond with proactive, context-aware strategies.

To facilitate the rapid formation of vehicle platoons through merging sections with significant curvature differences, an innovative game-theoretic method that combines the principle of social optimality with level-k game theory is proposed. In the design of the payoff function within game-theoretic models, potential field theory is introduced to quantify the potential safety risks of vehicles. The potential field function has the advantage of encapsulating the equivalent mass of the vehicle, velocity, acceleration, and other state information, allowing it to adapt to a heterogeneous vehicle environment. By adjusting its shape, the potential field function can reflect different driving styles [32]. The merging point is designated the epicenter of an attractive potential field, which coordinates vehicle movement toward targets and maintains stable spacing. Upon completing the merging process, the vehicles form a new platoon that maintains stable operation, thereby preventing secondary conflicts within the roundabout.

Upon entry into a roundabout, vehicles share information via roadside tactical controller (RTC) or vehicle-to-vehicle (V2V) communication to evaluate potential future trajectory intersections that may lead to conflicts [33]. After assessing collision risk, the entering vehicle evaluates the projected cooperative payoffs for both itself and the obstacle vehicle under two modalities: assuming the role of a leader versus that of a follower. Vehicles are projected onto the trajectories of each other, creating a virtual platoon. The Frenet coordinate system is employed to decouple the lateral and longitudinal dynamics of the vehicles, and a multi-vehicle error model is established. The vehicle entering the roundabout is designated the ego vehicle, with the vehicles already present within the roundabout categorized as obstacle vehicles. Noted that the obstacle vehicles could be either human-driven vehicles (HDVs) or CAVs. The ego vehicle utilizes the historical behavior of the obstacle vehicle to update the posterior probabilities associated with the reasoning level, and adapts its own reasoning level accordingly.

The main theoretical contributions of this study are summarized as follows:

(1) A novel and efficient merging strategy for vehicle platoons is proposed for roundabouts. The strategy is built on the integration of social optimality and level-k game theory, with all vehicles being projected into a one-dimensional virtual platoon through coordinate transformation to enable dynamic merging gap allocation.

(2) Based on the potential field function, a game-theoretic payoff function is designed to account for personalized driving styles. Additionally, an attractive force is applied at the merging point, with the purpose of propelling vehicles toward a harmonious merging process.

(3) To accurately predict the motion states of the obstacle vehicle, the ego vehicle updates the k-level probability of the obstacle vehicle using Bayesian probability formulas. Moreover, it employs an adaptive level adjustment mechanism to prevent deadlock between agents of the same level.

The remainder of this article is outlined as follows. Section II presents the formulations of the vehicle dynamics and multivehicle error model in the Frenet coordinate system. In Section III, the reward function of the game-theoretic framework is designed from dimensions such as merging efficiency, merging point location, safety, and ride comfort; subsequently, the entire game-theoretic framework is established. Section IV focuses on predicting the motion states of obstacle vehicles under different levels of intelligence and presents the Bayesian probability update process. Section V introduces the simulation and experimental setup, examines the stability of vehicle platoons, and compares the results of multi-vehicle merging under different control strategies, verifying the effectiveness and feasibility of the control strategy. Finally, Section 6 provides a comprehensive summary and discussion.

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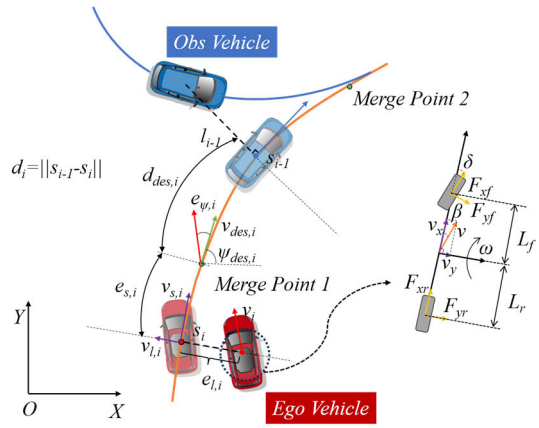

Fig. 1. Vehicle dynamics modeling in the Frenet coordinate system.

II. Multi-Vehicle Modeling in the Frenet Coordinate System

A. Lateral Control

The lateral and longitudinal dynamics modes of vehicles are coupled in the traditional Cartesian coordinate system, which is unsuitable for roads with variable curvature [34]. By constructing a Frenet coordinate system based on a prescribed reference trajectory, the longitudinal displacement s and the lateral offset l of a vehicle are decoupled, as shown in Fig. 1. Simultaneously, the obstacle vehicles are likewise projected onto the reference trajectory to constitute a virtual platoon. This transformation simplifies the complex two-dimensional merging problem into a one-dimensional following problem. However, the selection of the optimal merging point from two available options necessitates the application of game-theoretic methods to determine the most beneficial position.

As shown in (1), a two-degree-of-freedom (2)-DOF) vehicle dynamics model is constructed in the Cartesian coordinate system. In this case, due to the significant variations in the curvature along the roundabout, it is necessary to consider the possibility of tire side-slip and drift. The magic formula tire model is employed in (2) to describe the nonlinear relationship between the tire lateral force and slip angle.

$$\begin{cases} \dot{v}_{x,i} = \dot{\psi}_i v_{y,i} + a_i \\ \dot{v}_{y,i} = -\dot{\psi}_i \dot{x}_i + \dfrac{2}{m_i}\left(F_{yf,i}\cos\delta_{f,i} + F_{yr,i}\right) \\ \ddot{\psi}_i = \dfrac{2}{I_{z,i}}\left(L_{f,i}F_{yf,i} - L_{r,i}F_{yr,i}\right) \end{cases} \tag{1}$$

$$F_{y^*,i} = -D_* \sin\left\{C_* \tan^{-1}\left[B_*\alpha_* - E_*\left(B_*\alpha_* - \tan^{-1}(B_*\alpha_*)\right)\right]\right\} \tag{2}$$

where $v_{x,i}$ and $v_{y,i}$ are the longitudinal speed and lateral speed of vehicle i, ψ is the yaw angle, β is the slip angle, a_i is the acceleration, x_i is the longitudinal displacement, $F_{yf,i}$ and $F_{yr,i}$ are the lateral forces of the front wheels and rear wheels, $C_{af,i}$ and $C_{ar,i}$ are the cornering stiffnesses of the front wheels and rear wheels, $\delta_{f,i}$ is the steering angle of the front wheel, m_i is the mass of the vehicle body, $I_{z,i}$ is the moment of inertia of the vehicle body, and $L_{f,i}$ and $L_{r,i}$ are the front axis distance and rear axis distance to the center of mass, respectively. $* \in \{f, r\}$, D_* is the peak factor, B_* is the stiffness factor, C_* is the curve shape factor, E_* is the curvature factor, and α_* is the side-slip angle. The values of D_*, B_*, C_* and E_* can be obtained through experimentation, while α_* can be approximated using (3).

$$\begin{cases} \alpha_{f,i} = -\delta_{f,i} + \dfrac{v_{y,i} + L_{f,i}\dot{\psi}_{d,i}}{v_{x,i}} \\ \alpha_{r,i} = \dfrac{v_{y,i} - L_{r,i}\dot{\psi}_{d,i}}{v_{x,i}} \end{cases} \tag{3}$$

The transformation between Cartesian coordinates and Frenet coordinates is performed using (4). In practical computations and road applications, the reference trajectory is often composed of a series of discrete road points. First, the road point m closest to the projected point r of the vehicle is located, and then the vehicle state variables $[x, y, v_x, v_y, a_x, a_y, \psi]$ are transformed into Frenet coordinates via vector-based computation.

$$\begin{cases} s = s_m + \left|\left(\vec{d}_m \cdot \vec{\tau}_m\right)\vec{\tau}_m\right| \\ l = \left(\vec{r}_h - \vec{r}_r\right)\cdot \vec{n}_r \\ \dot{s} = v_s = \dfrac{\vec{v}_h \cdot \vec{\tau}_r}{1 - k_r l} \\ \dot{l} = v_l = \vec{v}_h \cdot \vec{n}_r \\ \ddot{s} = a_s = \dfrac{\vec{a}_h \cdot \vec{\tau}_h}{1 - k_r l} + \dfrac{\dot{s}^2 k_r l'}{1 - k_r l} + \dfrac{\dot{s}^2(k_r' l + k_r l')}{1 - k_r l} \\ \ddot{l} = a_l = \vec{a}_h \cdot \vec{n}_r - k_r(1 - k_r l)\dot{s}^2 \\ l' = \dfrac{\dot{l}}{\dot{s}} \\ l'' = \dfrac{\ddot{l} - l'\ddot{s}}{\dot{s}^2} \end{cases}$$

$$where \begin{cases} \vec{r}_h = [\vec{x}, \vec{y}] \\ \vec{v}_h = [\vec{v}_x, \vec{v}_y] \\ \vec{a}_h = [\vec{a}_x, \vec{a}_y] \\ \vec{\tau}_m = [\cos\theta_m, \sin\theta_m] \\ \vec{n}_m = [-\sin\theta_m, \cos\theta_m] \\ \vec{\tau}_r = [\cos\theta_r, \sin\theta_r] \\ \vec{n}_r = [-\sin\theta_r, \cos\theta_r] \end{cases} \tag{4}$$

where $\vec{\tau}_r$ and $\vec{n}_r$ are the tangent and normal vectors, respectively, to the projected point of the vehicle's center of mass on the reference trajectory; $\vec{\tau}_m$ and $\vec{n}_m$ are the normal and tangent vectors, respectively, of the nearest waypoint on the reference trajectory to the projection point; θ is the heading angle; s is the longitudinal displacement in the Frenet coordinate system; and l is the lateral displacement.

The multivehicle following error is decomposed into two parts: the longitudinal error relative to the virtual projection of the leading vehicle on the same trajectory and the lateral error of the vehicle itself relative to the projected trajectory. The relative relationship between the projection on the reference trajectory in Fig. 1 and the actual position of the vehicle can be determined:

$$\begin{cases} e_{l,i} = \left|\vec{x} - \vec{x}_m\right| \cdot \vec{n}_m \\ \dot{e}_{l,i} = v_{x,i}\sin e_{\psi,i} + v_{y,i}\cos e_{\psi,i} \\ e_{\psi,i} = \psi_i - \theta_r = \psi_i - (\theta_m + k_r \cdot \left|\vec{x} - \vec{x}_m\right| \cdot \vec{\tau}_m) \\ \dot{e}_{\psi,i} = \dot{\psi}_i - \dfrac{v_{x,i}\cos e_{\psi,i} - v_{y,i}\sin e_{\psi,i}}{1 - k_r e_{\psi,i}} \end{cases} \tag{5}$$

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$$\begin{bmatrix} \dot{e}_{l,i} \\ \ddot{e}_{l,i} \\ \dot{e}_{\psi,i} \\ \ddot{e}_{\psi,i} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & \frac{A}{m_i v_{x,i}} & -\frac{A}{m_i} & \frac{B}{m_i v_{x,i}} \\ 0 & 0 & 0 & 1 \\ 0 & \frac{B}{I_{z,i} v_{x,i}} & -\frac{B}{I_{z,i}} & \frac{C}{I_{z,i} v_{x,i}} \end{bmatrix} \begin{bmatrix} e_{l,i} \\ \dot{e}_{l,i} \\ e_{\psi,i} \\ \dot{e}_{\psi,i} \end{bmatrix} + \begin{bmatrix} 0 \\ -\frac{C_{af,i}}{m_i} \\ 0 \\ -\frac{L_{f,i} C_{af,i}}{I_{z,i}} \end{bmatrix} \delta_{f,i}$$

$$A = C_{af,i} + C_{ar,i}, B = L_{f,i} C_{af,i} - L_r C_{ar,i},$$
$$C = L_{f,i}^2 C_{af,i} + L_r^2 C_{ar,i} \tag{6}$$

where $e_{l,i}$ is the lateral error, $e_{\psi,i}$ is the heading angle error, u_i is the longitudinal control input of vehicle i, u_{i-1} is the longitudinal control input of vehicle *i-1*, and τ is the time constant. The lateral tracking errors of vehicle $e_{y,i}$ and $e_{\psi,i}$ are controlled by a feedforward linear quadratic regulator (LQR) controller [35].

B. Longitudinal Control

In Frenet coordinates, the relative longitudinal distance error e_s between vehicles can be expressed as:

$$\begin{cases} e_{s,i} = s_{i-1} - s_i - L - d_{safe} - h_i \cdot e_{v,i}, \ \ if \ \ s_{i-1} > s_i \\ e_{s,i} = s_{i-1} + L + d_{safe} + h_i \cdot e_{v,i} - s_i, \ \ if \ \ s_i > s_{i-1} \end{cases} \tag{7}$$

where d_{safe} is the safe intervehicle spacing. h is the headway time.

Meanwhile, the velocity error and acceleration error can be obtained:

$$e_{v,i} = \dot{e}_{s,i} = v_{s,i-1} - v_{s,i} - h a_{s,i} \tag{8}$$
$$e_{a,i} = \dot{e}_{v,i} = a_{s,i-1} - a_{s,i} - h \dot{a}_{s,i} \tag{9}$$
$$\dot{e}_{a,i} = \dot{a}_{s,i-1} - \dot{a}_{s,i} - h_i \ddot{a}_i = -\frac{1}{\tau} e_{a,i} + \frac{1}{\tau} u_{i-1} - \frac{1}{\tau} q_i \tag{10}$$
$$q_i = h_i \dot{u}_i + u_i \tag{11}$$

where, $e_{v,i}$ is the speed error, and $e_{a,i}$ the acceleration error. q_i as input for the system serves to stabilize the dynamic error and compensate for the input u_{i-1}. According to the PD control law, q_i can be designed as:

$$q_i = k_p e_{s,i} + k_d e_{v,i} + u_{i-1} \tag{12}$$
$$\dot{u}_i = -\frac{1}{h_i} u_i + \frac{1}{h_i} \left(k_p e_{s,i} + k_d e_{v,i} + u_{i-1} \right) \tag{13}$$
$$\begin{bmatrix} \dot{e}_{s,i} \\ \dot{e}_{v,i} \\ \dot{e}_{a,i} \\ \dot{u}_i \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -\frac{k_p}{\tau} & -\frac{k_d}{\tau} & -\frac{1}{\tau} & 0 \\ -\frac{k_p}{h_i} & -\frac{k_d}{h_i} & 0 & -\frac{1}{h_i} \end{bmatrix} \begin{bmatrix} e_{s,i} \\ e_{v,i} \\ e_{a,i} \\ u_i \end{bmatrix} + \frac{1}{h_i} \begin{bmatrix} 0 \\ 0 \\ 0 \\ u_{i-1} \end{bmatrix} \tag{14}$$

where u_i is the input quantity. The values of u_i and u_{i-1} are determined through game theory in Section III. k_p, k_d are the control parameters of the feedback control law. Applying the Routh-Hurwitz stability criterion while using the fact that the state matrix in (14) is lower block-triangular. When k_p, $k_d > 0$, the system possesses stability. Finally, the vehicle acceleration in the Frenet coordinate system is transformed into the vehicle acceleration in the Cartesian coordinate system by using the inverse transformation of the Frenet coordinate system.

$$\begin{cases} \vec{r}_r = [\vec{x}_r, \vec{y}_r] \\ \vec{r}_h = \vec{r}_h + l\vec{n}_r \\ \left|\vec{v}_h\right| = \left[\dot{l}^2 + (1 - k_r l)^2 \dot{s}^2\right]^{1/2} \\ \left|\vec{a}_h\right| = \left\{ \begin{array}{l} \left[\dot{l} + k_r (1 - k_r l) \dot{s}^2\right]^2 + \\ \left[\ddot{s} (1 - k_r l) - \dot{s}^2 k_r l' + \dot{s}^2 \left(k_r' l + k_r l'\right)\right]^2 \end{array} \right\}^{1/2} \end{cases} \tag{15}$$

III. Vehicle Interactive Decision-Making Based on Game Theory

A. Payoff Function

At unsignalized roundabouts, vehicles typically rely on right-of-way rules to resolve conflicts upon entry, which in another way means assigning a leader-follower relationship [36]. Generally, vehicles need yield to circulating traffic already within the roundabout. However, the cautious driving style of autonomous vehicles often leads them to decelerate and yield right of way well in advance, even if they can pass through the intersection safely and efficiently. This over-courteous behavior, whether inside or outside the roundabout, could result in unnecessary traffic congestion. Therefore, it is necessary to optimize the decision-making algorithm to balance safety with smooth traffic flow.

In the context of vehicle platoon merging at roundabout intersections, the primary control objective encompasses the strategic planning of optimal longitudinal speed trajectories and the judicious selection of merging points. After the merge, the overarching goal is to ensure safe intervehicle spacing while operating at the desired speed. For vehicle $i, i-1 \in N$, the cooperative objective can be represented as:

$$\lim_{t \to \infty} \left\{ \left\| s_i - s_{des,i} \right\|, \left\| \dot{s}_i - \dot{s}_{des,i} \right\|, \|\ddot{s}_i\| \right\} = 0 \tag{16}$$

where $s_{des,i}$ is the desired intervehicle distance and $s_{des,i}$ is the desired speed of vehicle i.

Based on the above control objectives, the vehicle platoon merging problem is transformed into a multivehicle game problem, where the objective of the game is to minimize the total vehicle platoon payoff function to achieve the social optimum.

$$\begin{cases} G = (N, \ A, \ \mathcal{J}) \\ N = \{1, 2, \cdots, n\}; A = \prod_{i=1}^{n} \pi_i; \mathcal{J} = \min \sum_{i=1}^{n} J_i \end{cases} \tag{17}$$

Where N is the set of vehicles involved in the game, A is the set of vehicle actions, π_i is the strategy set of vehicles i, and J is the set of payoff functions. The benefits of assigning different roles to vehicles within the merging context are assessed based on five critical indicators: the merging security risk, merging distance error, merging efficiency, smoothness, and desired speed error.

1) Merging Security Risk: In the merging process, the paramount objective is to ensure safety. The artificial potential field, utilized as a quantitative measure for latent safety hazards, has been extensively applied in the realm of CAVs [37]. Following numerous iterations of optimization, the improved

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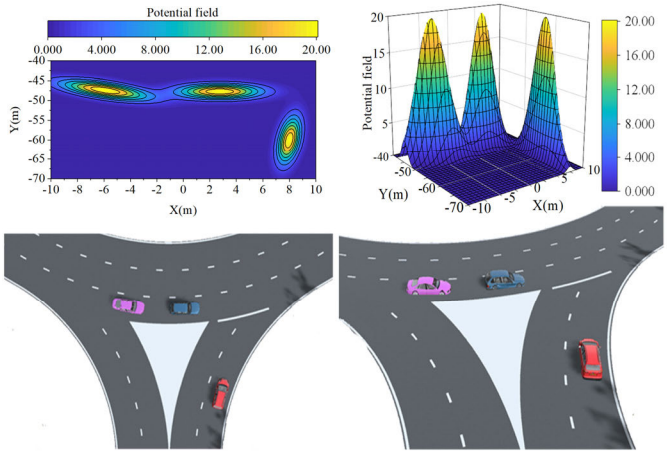

Fig. 2. Potential driving risk at a roundabout intersection.

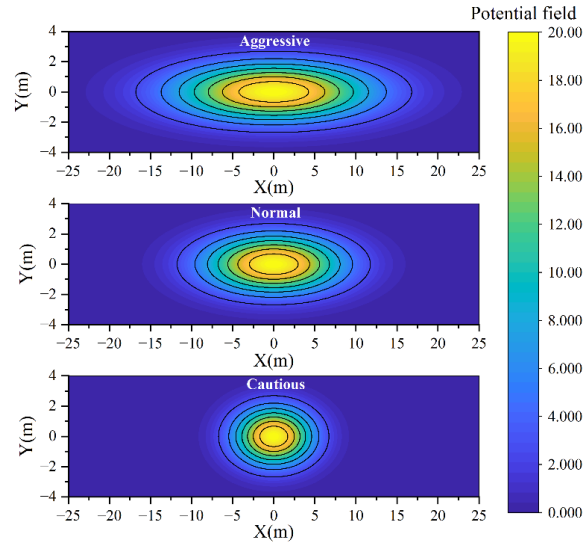

Fig. 3. The artificial potential field corresponding to different driving styles (Aggressive: x_σ = 25 m, Normal: x_σ = 17.5 m; Cautious: x_σ = 10 m).

potential field could reflect the safety risks posed by vehicles with different masses, velocities and accelerations (see Fig. 2).

As shown in (18), the artificial potential field function built based on a two-dimensional Gaussian distribution is given:

$$w_{s,i} = A_o e^{-\left(a_\varphi((s_{i-1}-s_i)/\delta)^2 + b_\varphi((s_{i-1}-s_i)/\delta)(l_{i-1}-l_i) + c_\varphi (l_{i-1}-l_i)^2\right)} \tag{18}$$

$$\delta = \frac{M_{i-1}}{m_{i-1}} \times \begin{cases} e^{(-|s_{i-1}-s_i|)} + p + a_{i-1}, & a_{i-1} > 0 \wedge s_{i-1} \geq s_i \\ \left(e^{(-|s_{i-1}-s_i|)} + 1\right)/5, & a_{i-1} > 0 \wedge s_{i-1} \geq s_i \\ \left(e^{(-|s_{i-1}-s_i|)} + 1\right)/5, & a_{i-1} < 0 \wedge s_{i-1} \geq s_i \\ e^{(-|s_{i-1}-s_i|)} + p + |a_{i-1}|, & a_{i-1} < 0 \wedge s_{i-1} \leq s_i \\ 1 & a_{i-1} = 0 \end{cases} \tag{19}$$

$$M_{i-1} = m_{i-1} \cdot (1.566 \times 10^{-14} \cdot v_{i-1}^{6.687} + 0.3345) \tag{20}$$

where A_o is the amplitude of the obstacle potential field, (s_j, l_j) are the obstacle coordinates, p is the risk factor adjustment parameter, and M_{i-1} is the equivalent mass of the obstacle vehicle. m_{i-1} is the mass of the obstacle vehicle. $a_{\varphi,i}$, $b_{\varphi,i}$, and $c_{\varphi,i}$ are the potential field rotation parameters, which can be calculated by the following equation:

$$\begin{aligned} a_{\psi,i} &= \frac{\cos^2\psi_i}{2\sigma_{x,i}^2} + \frac{\sin^2\psi_i}{2\sigma_{y,i}^2} \\ b_{\psi,i} &= -\left(-\frac{\sin(2\psi_i)}{4\sigma_x^2} + \frac{\sin(2\psi_i)}{4\sigma_y^2}\right) \\ c_{\psi,i} &= \frac{\sin^2\psi_i}{2\sigma_{x,i}^2} + \frac{\cos^2\psi_i}{2\sigma_{y,i}^2} \end{aligned} \tag{21}$$

The standard deviations in the longitudinal and lateral directions can be derived from the Gaussian equation without rotation:

$$\begin{aligned} \sigma_{x,i} &= \sqrt{-\frac{x_\sigma^2}{2\ln(0.01)}} \\ \sigma_{y,i} &= \sqrt{-\frac{y_\sigma^2}{2\ln(0.01)}} \end{aligned} \tag{22}$$

x_σ and y_σ determine the extent of influence of the potential field. Different driving styles are characterized by varying degrees of attention given to safety. According to reference, driving styles are categorized as aggressive, normal or cautious. This classification is instrumental in adjusting the longitudinal impact range of the potential field, as shown in Fig. 3 [38]. In subsequent games, the extent of the risk potential field will impact the ego vehicle's evaluation of the advantages associated with altering its own merging order. This will consequently determine whether the ego vehicle assumes the role of the leader or follower in relation to the obstacle vehicle.

2) Merging Distance Error: Each vehicle is associated with two prospective merging points. When the ego vehicle assumes the role of the leader, the merging point is positioned in front of the obstacle vehicle. In another situation, when the ego vehicle is designated as the follower, the merging point lies behind the obstacle vehicle. The reward increases commensurately as the vehicle approaches the specified merging point. This behavior is simulated by an attractive artificial potential field model. The reward for the merging distance error can be articulated as follows:

$$w_g = A_g\left(1 - e^{-a_{\varphi g}(s_i - s_{des,i})^2}\right) \tag{23}$$

where A_g is the amplitude of the gravitational potential field and $a_{\varphi g}$ is the potential field shape parameter.

3) Merging Efficiency: To ensure that vehicles participating in the merging process promptly reach the merging point and complete the merging maneuver with minimized velocity loss, the efficiency return for merging is defined as follows:

$$\begin{cases} w_e = k_v(v_{s,i} - v_{m,des})^2 - k_m\left(\dfrac{(v_{s,i} + v_{m,des})}{2d_m}\right)^2 \\ v_{m,des} = \dfrac{(s_j - s_i + d_{safe})}{t_h} \end{cases} \tag{24}$$

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where d_m is the distance to the merging point; when the vehicle exceeds the merging point, $d_m = +\infty$. $v_{m.des}$ is the desired speed of the merging process. t_h is the headway time. k_v is the weighting factor for tracking the desired speed. k_m is the weighting factor for the efficiency of arriving at the merging point.

4) Ride Comfort: Ride comfort is usually related to the increase in acceleration. In the present study, the ride comfort payoff function is defined as a quadratic term function of the vehicle acceleration and acceleration increment.

$$w_a = a_{s,i}^2 + \Delta a_{s,i}^2 \tag{25}$$

5) Desired Speed Error: The expected speed benefit primarily serves to facilitate the convergence of post merging vehicle platoon speeds:

$$w_v = A_v e^{a_{\varphi g}\left(v_{s,i} - v_{des,i}\right)^2} \tag{26}$$

where A_v is the amplitude of the desired speed error. When $d_m = +\infty$, $A_v = A_g$; otherwise, $A_v = 0$.

Combining (18) to (26), the overall payoff function J_i of vehicle i is

$$\begin{aligned} &J_i\left(s_i, s_j, v_{s,i}, v_{s,j}, a_{s,i}, a_{s,j}, \Delta a_{s,i}\right) \\ &= q_s w_s + q_g w_g + q_e w_e + q_a w_a + q_v w_v \end{aligned} \tag{27}$$

where q_s, q_g, q_e, q_a and q_v are the weighting parameters. Based on the principle of social optimality, each vehicle adopts action strategies π_i^* within an Np-step prediction time horizon. γ is the time decay coefficient for the payoff function. The Nash equilibrium of the game is determined by employing a receding horizon optimization and distributed computing to iteratively approach the solution.

$$\begin{aligned} &\mathcal{J}^*\left(\pi_1^*, \cdots, \quad \pi_i^*, \cdots, \pi_n^*\right) \le \mathcal{J}\left(\pi_1, \cdots, \pi_i, \cdots \pi_n\right) \\ &\mathcal{J}\left(\pi_1, \cdots, \pi_i, \cdots \pi_n\right) = \sum_{i=1}^{n} J_i\left(\pi_i\right) \\ &\pi_i = \left\{a_{s,i}(1), \cdots, a_{s,i}(t), \cdots a_{s,i}(Np)\right\} \\ &\begin{cases} \min\limits_{\pi_i^* \in A^*} \sum\limits_{t=1}^{Np} \gamma^t J_i\left(s_i(t), s_j(t), v_{s,i}(t), v_{s,j}(t), a_{s,i}(t), a_{s,j}(t)\right) \\ \text{s.t. } \xi_i(t+1) = E\xi_i(t) + F\Delta a_{s,i}(t) \\ v_{s,i}(t) < v_{\max} \\ -a_{\max} \le a_{s,i}(t) \le a_{\max} \\ -\Delta a_{\max} \le \Delta a_{s,i}(t) \le \Delta a_{\max} \\ \left|s_i(t) - s_j(t)\right| \ge d_{safe} + L \end{cases} \end{aligned} \tag{28}$$

Where, ξ_i is the state parameters matrix of the vehicle i. In the Frenet coordinate system, the vehicle state ξ_i is updated based on the vehicle kinematic model, as expressed in (29).

$$\xi_i(t+1) = E\xi_i(t) + F\Delta a_{s,i}(t) \tag{29}$$

$$\xi_i(t) = \begin{bmatrix} s(t) \\ v_s(t) \\ a_s(t) \end{bmatrix}, E = \begin{bmatrix} 1 & \tau & 0.5\tau^2 \\ 0 & 1 & \tau \\ 0 & 0 & 1 \end{bmatrix}, F = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \tag{30}$$

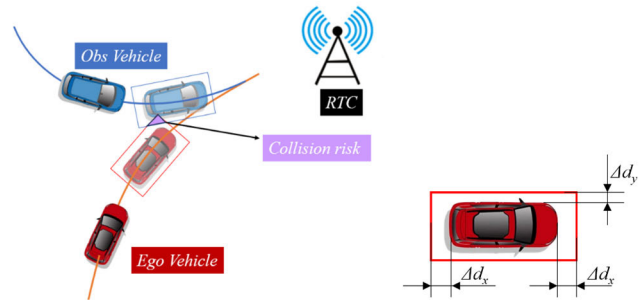

Fig. 4. RTC device detects the collision risk.

B. Level-k Game

The beginning of the level-k game necessitates the fulfillment of two conditions. Firstly, both the ego vehicle and the obstacle vehicle must concurrently enter within a 50 m radius of the roundabout intersection. Secondly, the RTC identifies a potential collision risk between the two vehicles within the ensuing 5 s, as shown in Fig. 4. In this study, the RTC identifies potential collision hazards by detecting whether there is an intersection of the vehicles' safety margin zones in their spatiotemporal trajectories. The dimensions of the vehicles' safety margin zone are defined as follows: the distance from the rectangle to the front and rear bumpers of the vehicle, $\Delta d_x = (0.5L + 0.1v)$, while the distance from the rectangle to the side edges of the vehicle body, $\Delta d_y = 0.5W$. The vehicles predict their future trajectories for the next 5 s recursively in their own Frenet coordinate systems using the kinematic model in (30).

In real-world game-theoretic interactions, players with intelligent reasoning capabilities (e.g., human drivers and CAVs) perform limited reasoning about the behavior of other players to maximize their own benefits [39]. Generally, level- *0* agents exhibit selfish behavior, as they consider other agents in their vicinity to be stationary. Furthermore, an agent of level-k ($k \ge 1$) assumes that other agents are operating at level-(k-*1*) intelligence when conducting its strategic reasoning, which is shown in (31).

$$\begin{aligned} &\left(\pi_i^*\right)^k = \left\{\left(a_i^*(1)\right)^k, \left(a_i^*(t)\right)^k, \cdots, \left(a_i^*(Np)\right)^k\right\} \\ &\in \arg\min_{\pi_i^* \in A^*} \sum\nolimits_{t=1}^{Np} \gamma^t J_i\left(\left(\pi_1^*\right)^{k-1}, \cdots \left(\pi_i^*\right)^k, \cdots, \left(\pi_n^*\right)^{k-1}\right) \end{aligned} \tag{31}$$

Research indicates that the reasoning level of human drivers typically does not surpass level-*2* [40]. Furthermore, the drivers' reasoning levels, to a certain degree, reflect their pReferences in driving styles. For instance, an intelligent agent at level-*0* is inclined toward aggressive driving, an agent at level-*1* is inclined toward conservative driving, and an agent at level-*2* exhibits more rational and balanced driving behavior [41]. There are two purposes for employing level-k game-theoretic strategies at intersections. First, it aims to simulate the dynamic interplay with intelligent agents that possess diverse driving styles and reasoning capabilities, as would be encountered in realistic settings. Second, and more importantly, it integrates the anticipated reactions of other vehicles to the ego vehicle's strategic trajectory planning into the cognitive reasoning process. This integration unites the

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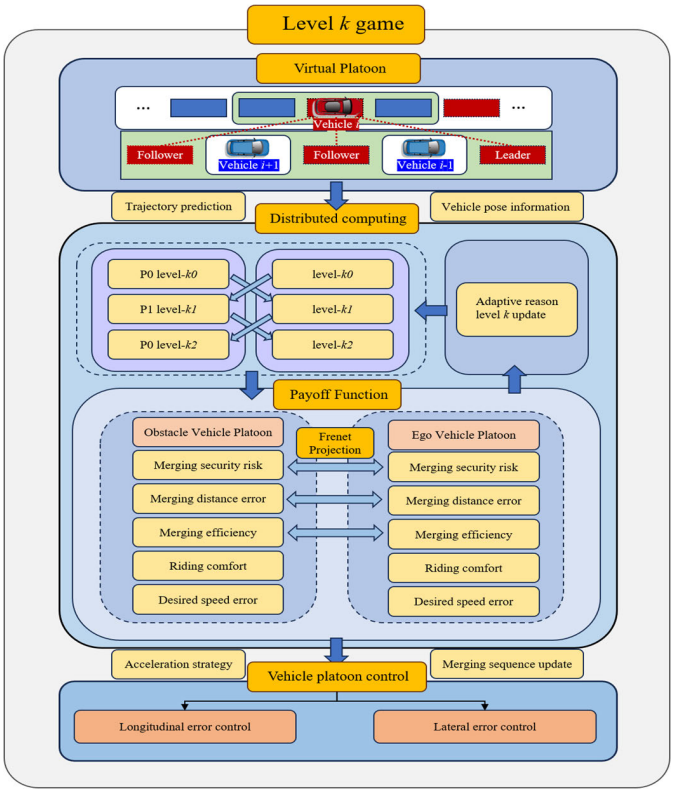

Fig. 5. Framework of the level-k game process.

predictive and planning modules, enabling CAVs to navigate with greater self-assurance driving strategies.

Generally, intelligent agents who share an equivalent level of reasoning are prone to entering conflicts. For instance, conflicts between two level-*0* or two level-*1* agents can often lead to collisions or traffic deadlocks. Conversely, interactions between agents of different reasoning levels are less likely to result in deadlocks, as the higher-level agent can anticipate the behavior of the lower-level agent. Nonetheless, relentlessly increasing the reasoning level in CAVs is not an impeccable strategy. An increase in the k value within the level-k game means an additional layer of strategic iteration, which increases the computational requirements in complex scenarios. Moreover, it can amplify the trajectory prediction error when forecasting the behaviors of others, potentially leading to suboptimal outcomes. Therefore, when encountering an obstacle vehicle with level-*2* intelligent decision-making, the ego vehicle can mitigate the risk of deadlock by downgrading its decision-making level to *k-1*. The framework of the whole level-k game process is plotted in Fig. 5.

After the vehicles complete the virtual platoon projection, vehicle i engages in a game with its two neighboring vehicles. For vehicle i, there are three potential merging points: to become the leading vehicle for vehicle $i-1$, the following vehicle for vehicle $i-1$, or the following vehicle for vehicle $i+1$. Based on the reasoning level estimation of the neighboring vehicles and the payoff function, vehicle i determines the optimal merging sequence and action strategy. This process updates the level-k iterative results. Through distributed computing, the system eventually reaches a socially optimal outcome. The lower controller then executes the longitudinal and lateral control of the vehicle platoon based on the prescribed acceleration strategy.

The game concludes when the final participating vehicle successfully passes the roundabout intersection. At this juncture, the unified vehicle platoon control assumes responsibility for the collective new platoon.

C. Adaptive Level-k Updating

The reasoning level of an intelligent agent significantly influences subsequent trajectory prediction and the selection of driving strategies. A mismatch between the level-k assumption and the actual performance of the agent can engender erroneous decisions and increased driving safety risk [42]. To mitigate this problem, the ego vehicle employs a Bayesian updating mechanism to adaptively estimate the reasoning levels of other vehicles. This estimation is predicated on the historical interactive behaviors of the obstacle vehicles.

According to the maximum likelihood principle:

$$\tilde{k}_p^j \in \arg\max_{k\in K} P^j\left(k^j = k|t\right) \tag{32}$$

where, k^j is the reasoning level of vehicle j and P^j represents the belief distribution of the different intelligence levels of vehicle j. According to the Bayesian probability formula in (33), the posterior probability $p^j(k^j = k|t)$ is:

$$P^j(k^j = k|t) = \frac{p^j(k^j = k|t-1)p^j(t-1|k^j = k)}{\sum_{k\in K} p^j(k^j = k|t-1)p^j(t-1|k^j = k)} \tag{33}$$

where, $p^j(t\text{-}1|k^j = k)$ represents the probability that the real acceleration a^j_{real} observed in vehicle j at time *t-1* matches the acceleration prescribed by the strategy $\pi^j(k^j{=}k|t\text{-}1)$ it adopts at reasoning level k, $K = \{0, 1, 2\}$. The probability distribution of vehicle acceleration estimated within the time series Np is assumed to follow a Gaussian distribution. It is anticipated that the deviation of the data will amplify with extended forecast durations. This results in a diminished probability of occurrence for those data points. Within the sequence of action strategies, the probability of executing each particular action is represented by $\{\zeta, \zeta^2, ..., \zeta^{Np}\}$:

$$\begin{aligned}&\pi^j(k^j = k|t)\\ &= \left\{a^j_{1|Np}(k^j = k|t), a^j_{2|Np}(k^j = k|t), \ldots, a^j_{Np|Np}(k^j = k|t)\right\}\end{aligned} \tag{34}$$

$$p(u^j(k^j = k|t)) = \left\{\zeta, \zeta^2, \ldots, \zeta^{Np}\right\} \tag{35}$$

where the mathematical expectation μ and standard deviation can be calculated by (36) and (37). When the actual acceleration of the obstacle vehicle is a^j_{real}, the probability estimate $p^i(t\text{-}1|k^j = k)$ is calculated by reference to the Gaussian cumulative distribution.

$$\mu = E\left(\pi^j\left(k^j = k \mid t\right) \cdot p^T\left(u^j\left(k^j = k \mid t\right)\right)\right) \tag{36}$$

$$\sigma^2 = \frac{\sum_{i=1}^{Np}\left(a^j_{i|Np}\left(k^j = k \mid t\right) - \mu\right)}{Np-1} \tag{37}$$

$$p^i(t-1|k^j = k) = \int_{a^j_{real}}^{+\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx \tag{38}$$

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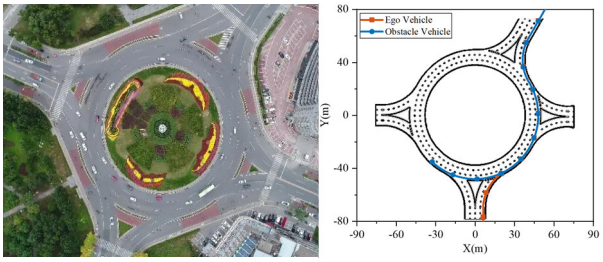

Fig. 6. The three-lane roundabout at the Yingbin Roundabout in Pinggu District, Beijing (left), and the modeled roundabout in the simulation scenario (right).

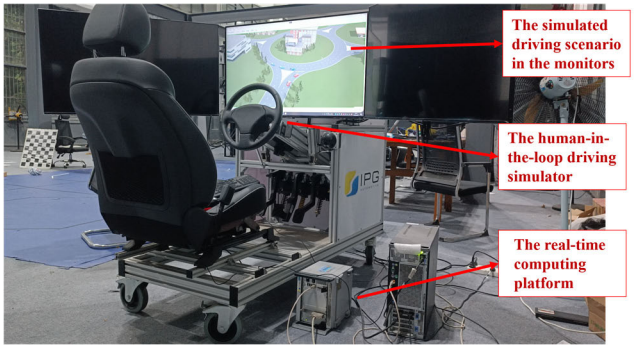

Fig. 7. The real-time hardware-in-the-loop platform.

IV. Roundabout Intersection Simulation

A. Roundabout Scenario

The simulation scenario for the roundabout experiments is modeled after the three-lane roundabout local at the Yingbin Roundabout in Pinggu District, Beijing. This roundabout has four entry and exit points, with an external radius of 50 m (as shown in Fig. 6). Moreover, vehicle conflict and merging issues at one of these entrances and exits, which serve as a representative case study, are selected. Among them, the red path is the route for ego vehicles to enter the roundabout. The blue path is the route that obstacle vehicles take when traveling around the roundabout.

Experiments were conducted using a real-time hardware-in-the-loop platform to assess the viability of the proposed algorithm. The platform is shown in Fig. 7, consists of three main components: a real-time computing platform, a human-in-the-loop driving simulator and a simulated driving scenario in the monitors. Among them, the real-time simulation module includes M410-Quadruple CAN/CAN FD Interface, F23P01 Quad-core i7 Single Board Computer (2.4GHz, 16GB RAM, and 6MB cache), F205 Carrier Board and Xpack4 RoadBox. The host computer in driving simulator configuration is 16 cores of CPUs (3.2GHz).

The HIL testing platform serves two primary functions. First, it provides an industry-standard vehicle dynamics simulation environment that incorporating realistic chassis parameters, tire models, and roadway scenarios. Second, it utilizes the real-time computing platform to validate the feasibility of the proposed algorithm. The real-time machine synchronizes simulation results with the control cycle of an actual vehicle by converting virtual scenario outputs into CAN bus signals, which are transmitted to the braking control unit and steering actuator within the HIL system. At the same time, ECU feedback is collected and integrated into the vehicle model, thereby emulating the operational conditions of the algorithm when deployed on a real vehicle. Following the one-to-one correspondence between a real-time unit and a HIL vehicle, the second following vehicle is selected as the HIL target in the platoon stability experiments, while the first vehicle in the ego vehicle platoon is selected in the game interaction experiments. The remaining vehicles are simulated and coordinated via network-based data exchange.

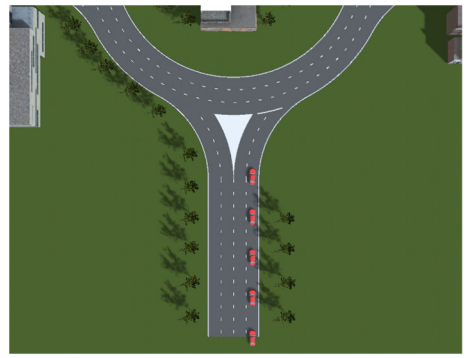

Fig. 8. Initial state of vehicle platoon at roundabout entry.

TABLE I
Vehicle Parameters

Parameter	Value	Unit
Vehicle length L_i	5.3	m
Vehicle mass m	2020	kg
Safe distance d_{safe}	3	m
Time constant τ	0.005	s
Time headway h	0.5	s
L_f	1.265	m
L_r	1.682	m
C_f	-175016	Nard^{-1}
C_r	-130634	Nard^{-1}
I_z	4095	kgm^2
[D_f, D_r]	[5868, 3975]	
[B_f, B_r]	[0.1932, 0.1942]	
[C_f, C_r]	[1.413, 1.413]	
[E_f, E_r]	[-0.2854, -0.2854]	

B. String Stability Analysis for Vehicle Platoon

To assess the stability of the vehicle platoon operating on the roundabout, a five-vehicle platoon configuration was established, as shown in Fig. 8. Each vehicle commenced with an initial speed of 30 km/h. The vehicle parameters are shown in Table I. The controller-related parameters are shown in Table II. At 5 s, Vehicle 1, acting as the leading vehicle, decelerated from 30 km/h to 18 km/h, and resumed its initial speed at the 10 s. This maneuver replicates the standing wave phenomenon observed in traffic flow. The state change of the remaining vehicles in the platoon is shown in Fig. 9, where the displacement, velocity, and acceleration are all defined along

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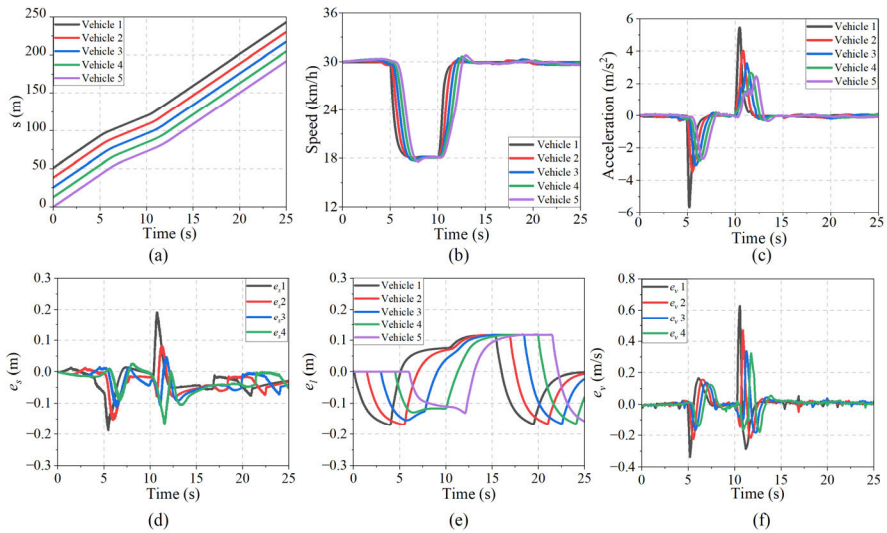

Fig. 9. Vehicle platoon stability experiment result (a) s position, (b) vehicle speed, (c) vehicle acceleration, (d) longitudinal error, (e) lateral error, (f) speed error.

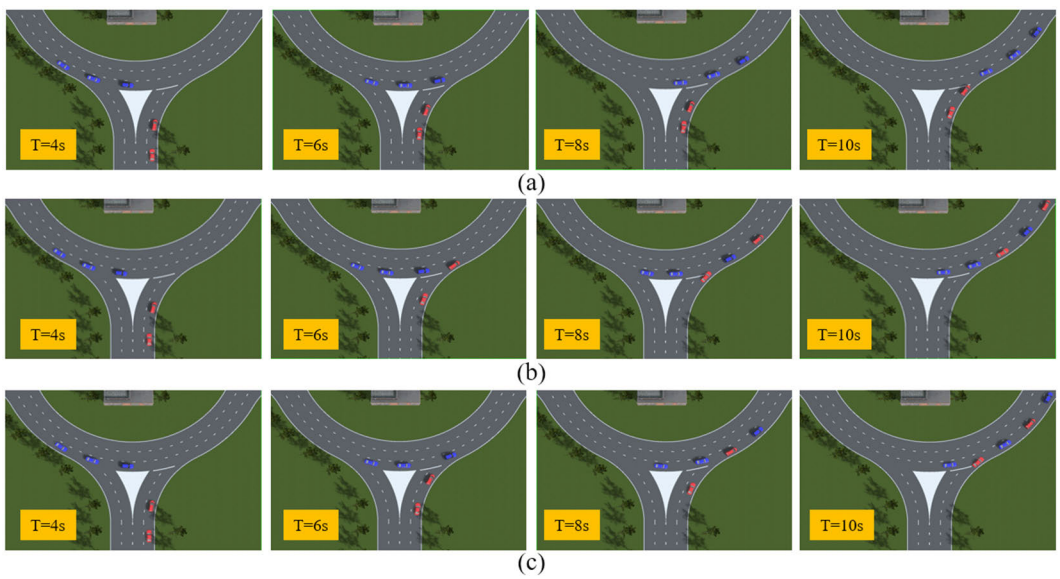

Fig. 10. (a) Interacting with a level-*0* obstacle vehicle platoon, (b) interacting with a level-*1* obstacle vehicle platoon, (c) interacting with a level-*2* obstacle vehicle platoon.

TABLE II
CONTROL PARAMETERS

Parameter	Value
A_o, A_g	20
p	0.5
y_σ	4
k_v	0.5
k_m	0.5
Np	40
$[q_s, q_g, q_e, q_a, q_v]$	[20, 200, 35, 80, 15]
$[\gamma, \zeta]$	[0.98, 0.95]

the tangential direction of each vehicle's global reference trajectory.

As shown in Fig. 9(a), the s-position represents the cumulative distance traveled along the road centerline. Throughout the acceleration and deceleration process, the platoon maintained stable behavior. The longitudinal following error remained within ± 0.2 m, while the lateral error was confined to a maximum of -0.17 m. The longitudinal speed error did not exceed 0.63 m/s. Both the longitudinal spacing error and speed error exhibit a clear attenuation trend, indicating effective disturbance suppression. Meanwhile, the acceleration amplitudes of all vehicles gradually decrease over time, further confirming the string-stable performance of the proposed platoon control strategy under roundabout driving conditions.

C. Merging of Two Vehicle Platoons

The merging process of ego vehicle platoon with obstacle vehicle platoon at differing reasoning levels into a new platoon is illustrated in Fig. 10. The scenario involves 3 vehicles navigating a roundabout, accompanied by 2 CAVs attempting

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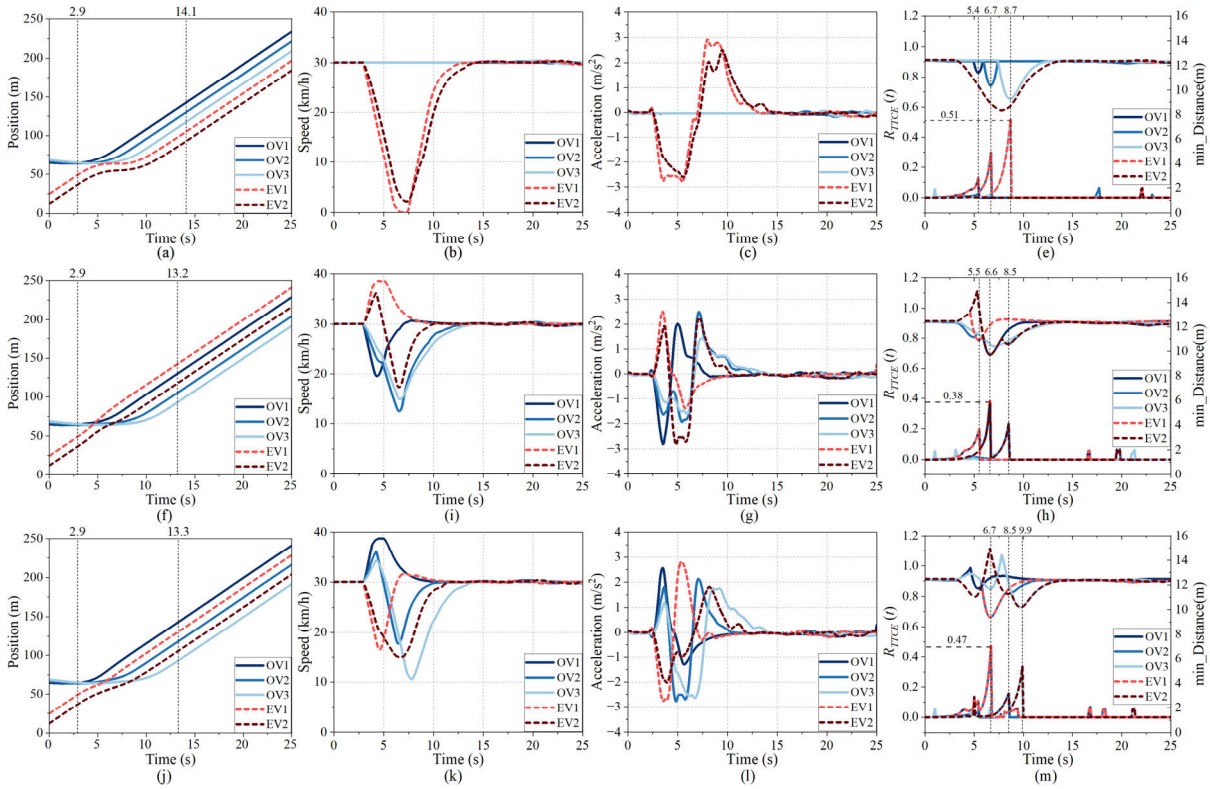

Fig. 11. The trajectories, speed, acceleration and TTCE of the vehicle platoon during the merging process (level-*0*: (a)-(e); level-*1*: (f)-(h); level-*2*: (j)-(m)).

to merging, with all vehicles satisfying the motion constraints of v_{max} = 40 km/h, a_{max} = 3 m/s^2, Δa_{max} = 5 m/s^3. At 2.9 s, the RSU identifies a potential collision risk between the two platoons and the initiates a level-*k* game to determine the optimal sequence of passage. For quantitative safety evaluation, the Time-to-Closest-Encounter (TTCE) metric is adopted [43]. TTCE is an enhanced version of the traditional Time-to-Collision (TTC), extending its applicability from one-dimensional lane-based scenarios to general two-dimensional motion environments. If the $R_{TTCE}(t) \geq 0.7$, a collision risk is deemed to exist.

Fig. 11 provides a detailed representation of the trajectories, velocities, accelerations, minimum Euclidean distance, and $R_{TTCE}(t)$ of the vehicles throughout the interaction process. When interacting with level-*0* obstacle vehicles (Ovs), the ego vehicles chose to decelerate and merge into the rear of the obstacle vehicle platoon, as shown in Fig. 11(a). The vertical axis of the trajectory plot represented the longitudinal displacement in the Frenet coordinate system, which was based on the path of the ego vehicle. At this reasoning level, the OVs did not respond to the ego vehicles' request to merge into the roundabout and maintained a steady speed of 30 km/h. Ego Vehicle 1 (EV1) stopped at 6.9 s with a peak deceleration of -2.76 m/s^2, while Ego Vehicle 2 (EV2) followed with a maximum deceleration of -2.60 m/s^2. The game concluded when EV2, the last vehicle in the platoon, passed the merging point at t = 14.1 s. Throughout the merging maneuver, EV1 and EV2 maintained an average speed of 18.0 km/h. The minimum Euclidean distance was recorded at 8.56 m, with v_{EV1} = 0.0 km/h and v_{EV2} = 2.0 km/h. This distance satisfied the minimum safe distance requirement $d_{safe} + L$ = 8.3 m. Additionally, as both vehicles were nearly stationary at that moment, the relative velocity was negligible, resulting in no significant collision risk. Peak values of the TTCE occurred three times, corresponding to EV1's yielding maneuvers for OV1, OV2, and OV3 at the roundabout entrance. The maximum recorded $R_{TTCE}(t)$ = 0.51, which remained within the pre-defined safety margin and indicated no immediate collision threat.

From the top-view perspective in Fig. 10, the new vehicle platoon is essentially complete at 10 s. This early completion reduces the pressure on subsequent platoon control and prevents significant acceleration fluctuations when the merged vehicles switch to platoon control.

When interacting with level-*1* obstacle vehicles, OV1 (Obstacle Vehicle 1) chose to yield, allowing EV1 to pass through the intersection first. Driven by the merging point ahead, EV1 accelerated to 38.8 km/h, quickly passed through the intersection, and took the lead in the platoon. Its speed then gradually decreased to the target speed of 30 km/h. OV1, acting as the follower of EV1, adjusted its speed to form a cooperative platoon with EV1. Meanwhile, the position of EV2 was projected between OV1 and OV2. Based on the game results, EV2 decelerated appropriately to follow OV1. This outcome was expected. If EV2 had tried to overtake OV1 to enter the intersection, OV1 would have experienced a larger speed reduction, negatively impacting the overall speed of the obstacle vehicle platoon and decreasing traffic efficiency. The game ended at 13.2 s, which was shorter than when interacting with the level-*0* obstacle vehicle platoon. This mainly

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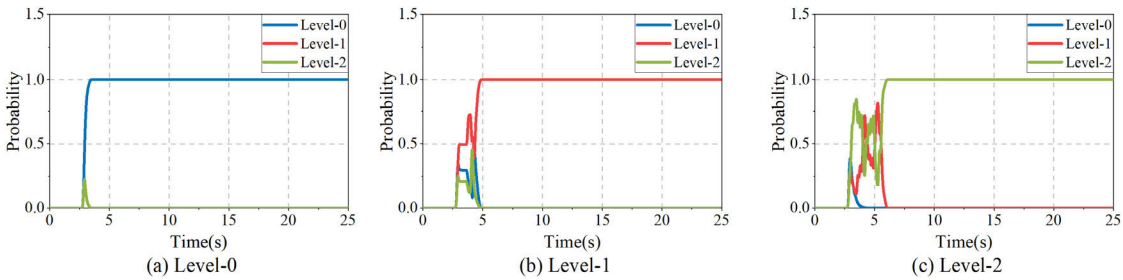

Fig. 12. Adaptive estimation of vehicle reason levels for obstacle vehicles.

resulted from EV1's acceleration during merging. Instead of maintaining a conservative speed, EV1 briefly accelerates and then quickly crosses the intersection. By cooperating with the obstacle vehicle platoon, it forms a new platoon, effectively reducing waiting time for subsequent vehicles. During the merging process, EV1's average speed was 32.4 km/h, while EV2's was 28.2 km/h. The minimum Euclidean distance was recorded at 9.72 m, with v_{OV1} = 28.9 km/h and v_{EV2} = 17.3 km/h. The maximum recorded $R_{TTCE}(t)$ = 0.38, was observed at this same moment, confirming the absence of a collision risk.

When interacting with level-*2* obstacle vehicles, the obstacle vehicle platoon initially saw the ego vehicle platoon as an opponent operating at reasoning level 1. This meant that the ego vehicle platoon would respond to the obstacle vehicle platoon's actions with a yielding reaction. As a result, the obstacle vehicle platoon planned its speed accordingly. OV1 attempted to accelerate and cross the intersection early, taking the lead within the platoon. In contrast, EV1 slowed down to yield, positioning itself behind OV1 based on its virtual projection position. OV2 strategically moved between EV1 and EV2. The game ended at 13.3 s. During the merging, EV1's average speed was 28.1 km/h, while EV2's was 23.8 km/h. The minimum Euclidean distance was recorded at 9.40 m, with v_{OV2} = 17.7 km/h and v_{EV2} = 15.1 km/h. The maximum recorded $R_{TTCE}(t)$ = 0.47, was observed at this same moment, confirming the absence of a collision risk.

Fig. 12 demonstrates the interactive process by which the ego vehicle ascertains the inference level of the obstacle vehicle. It was assumed that all vehicles within the same platoon shared the same reasoning level, thereby simplifying the scenario to focus solely on identifying the reasoning level of the leading vehicle of the platoon. The duration required for the ego vehicle to estimate the reasoning level of the obstacle vehicle lengthens with the escalation of the intelligent agent's reasoning level. Specifically, the estimation for a level-*0* agent is finalized in 3.4 s, for a level-*1* agent in 4.9 s, and for a level-*2* agent in 6.3 s. The primary reason for this is the inherent trajectory prediction ability of higher-level intelligent agents, which allows them to make similar behavioral strategy determinations within a specific predictive time interval. With an increase in the number of reasoning iterations, the magnification of prediction errors can also correspondingly affect the precision of Bayesian posterior probability updates.

When facing opponents of three different reasoning levels, the ego vehicle platoon safely and smoothly completed the merging process and maintained a stable platoon afterward. These results reduced the control pressure on subsequent platoon stability, demonstrating the effectiveness of the proposed level-*k* game-theoretic framework.

D. Comparison With the FIFO-VPS and MINLP Strategy

To demonstrate the improvement of the proposed game-theoretic model on the passage efficiency of vehicle platoons at roundabouts, comparisons were made with the FIFO-based virtual platooning (FIFO-VPS) and MINLP strategies. Among them, the FIFO-VPS projects all approaching on-ramp CAVs onto the mainline to form a virtual vehicle platoon, and assumes that all CAVs follow the principle of FIFO [44]. MINLP jointly optimizes the passage sequence at intersections and the vehicle trajectories, with the objective of minimizing the overall speed loss of the platoon [28].

The simulation scenario was configured as follows. Vehicle platoons, each consisting of 3 vehicles, were sequentially introduced into the roundabout from both the circulating lane and the incoming road. All platoons maintained an initial velocity of 30 km/h and an intra-vehicle spacing of 12.5 m. This process was repeated 5 times, with a 4 s interval between each iteration. The interactive obstacle vehicles were categorized into three distinct groups based on their reasoning capabilities: Level-*0* (non-cooperative), Level-*1* (altruistic yielding), and Level-*2* (rational response). Consequently, the ego vehicle platoons were required to adapt their merging strategies accordingly. Immediately after passing the merge point, each vehicle was governed by a homogeneous platoon controller. If the spacing error exceeds 3 m, the platoon would be disbanded and each car proceeded independently at the desired speed 30 km/h. Fig. 13 presents the position trajectory diagrams of the vehicles under the three merging strategies.

To evaluate the effectiveness of the proposed algorithm, a comparative analysis was conducted against the FIFO-VPS and MINLP algorithms across three key dimensions: safety, merging efficiency, and comfort. Specifically, safety was characterized by the maximum TTCE observed during the merging. Merging efficiency was assessed using the merging delay time, defined as the time lost by the merging vehicle due to interactions with mainline traffic.

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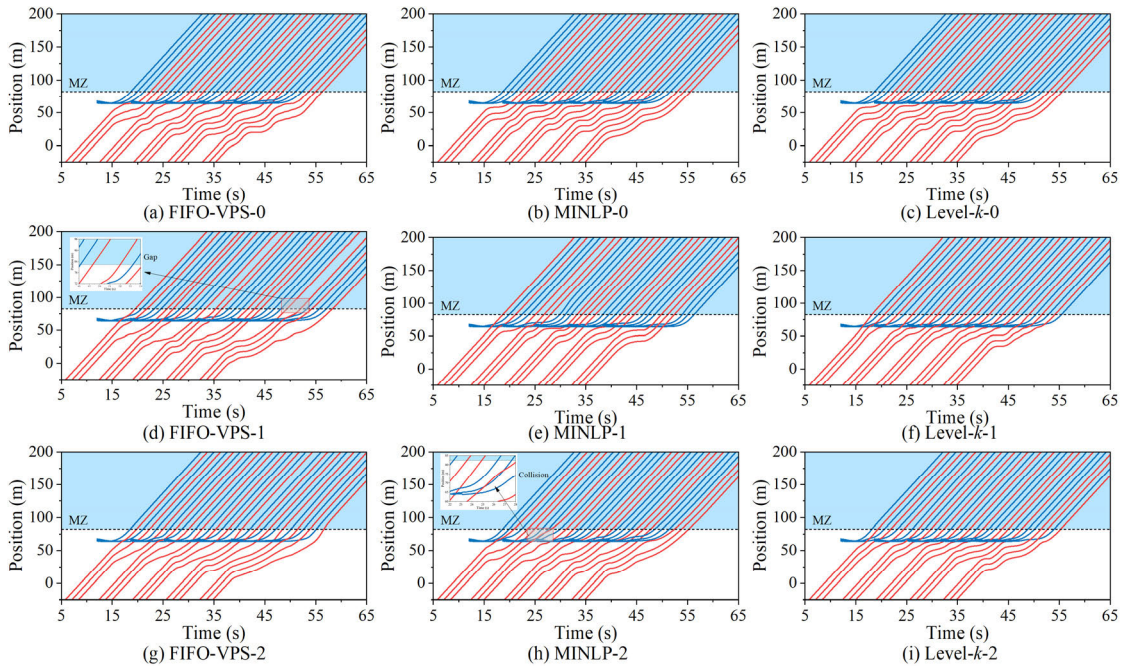

Fig. 13. Trajectory comparison of the three control strategies for vehicle platoons in high-density roundabout traffic. Columns (left to right): FIFO-VPS, MINLP, and the proposed level-k strategy. Rows (top to bottom): interactions with obstacle vehicles of reasoning levels 0, 1, and 2.

TABLE III
PERFORMANCE COMPARISON RESULTS

Indicators	Level-k			FIFO-VPS			MINLP		
OV reasoning level	0	1	2	0	1	2	0	1	2
Max.R_{TTCE}	0.51	0.38	0.47	**0.40**	0.62	**0.45**	0.68	**0.27**	0.98
Delay time (s)	**18.35**	**16.61**	**17.83**	18.95	21.05	19.30	18.85	18.60	18.69
Max.Longit.Accel. (m/s^2)	**2.17**	2.02	**2.04**	2.63	2.21	2.32	2.23	**1.88**	2.58
Max.Lat.Accel. (m/s^2)	**2.56**	2.51	**2.55**	3.59	2.75	2.61	2.77	**1.88**	3.43
Max.Longit.Jerk (m/s^2)	**0.89**	1.38	**1.15**	1.14	1.40	1.56	1.14	**1.23**	1.61
Max.Lat.Jerk (m/s^2)	**2.56**	3.17	3.52	2.82	3.85	3.77	3.15	**1.77**	**2.70**

Comfort was appraised through the maximum lateral acceleration, maximum longitudinal acceleration, maximum lateral jerk, and maximum longitudinal jerk experienced throughout the trajectory. A summary of these performance indicators is provided in Table III.

When encountering level-*0* obstacle vehicle platoons, all three strategies chose to decelerate and yield to the obstacle vehicle platoon, resulting in similar delays. Compared to the level-*k* strategy, the FIFO-VPS strategy did not account for the benefit of rapid platoon formation, resulting in slower integration of the new platoon. However, it achieved a lower TTCE value, indicating a safer margin. In contrast, the MINLP strategy continuously attempted to optimize the merging sequence during the process. However, this approach resulted in an elevated collision risk due to the obstacle vehicles' disregard for the merging intentions of the ego vehicles (EVs), ultimately demonstrating comparatively weaker overall robustness. Compared to FIFO-VPS and MINLP, the level-*k* game strategy demonstrated superior ride comfort. Furthermore, it achieved a 3.2% and 2.6% improvement in merging efficiency compared to the FIFO-VPS and MINLP strategies, respectively, when interacting with the level-*0* obstacle platoon.

When encountering level-*1* obstacle vehicle platoons, the deceleration-yielding behavior of obstacle vehicles amplified the shortcomings of FIFO-VPS. As shown in Fig. 14(a), conflicting vehicles decelerated mutually, leading to a deadlock situation. After attempting to restore the desired speed, the resulting inter-vehicle spacing within the original platoon became excessively large, ultimately causing the vehicles to disengage and creating undesirable gaps. In contrast, the MINLP strategy demonstrated superior ride comfort and achieved smooth merging. By prioritizing the minimization of overall speed loss for all vehicles, the ego platoon effectively utilized gaps created by the yielding behavior of surrounding vehicles. Although its efficiency was lower than that of the adaptive level-k game, it had the least negative impact on subsequent traffic flow. As vehicle density increased within the roundabout, a traffic pattern gradually emerged where the

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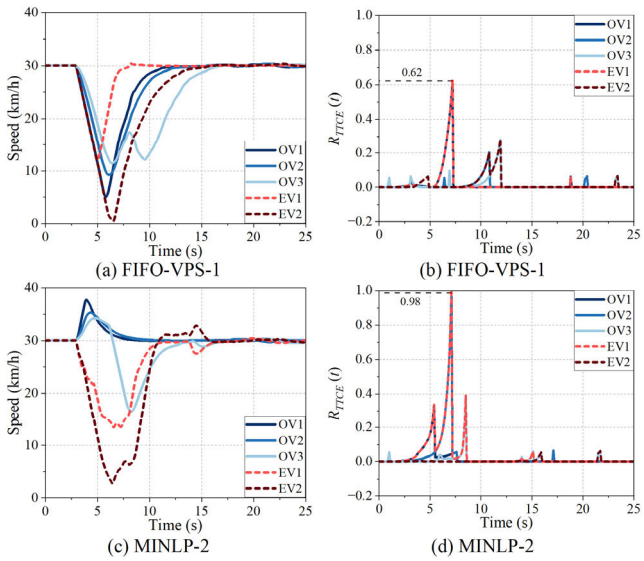

Fig. 14. (a) Speed profiles and (b) RTTCE profiles for the FIFO-VPS control strategy when interacting with Level-*1* obstacle vehicles; (c) Speed profiles and (d) RTTCE profiles for the MINLP control strategy under Level-*2* obstacle vehicles.

ego platoon passed through first, followed by the obstacle vehicle platoon. The proposed level-k strategy autonomously regulates vehicle speeds in response to merging opportunities, enabling it to effectively utilize available gaps and achieve the highest passing efficiency. This approach demonstrates efficiency improvements of 21.1% and 10.7% over the FIFO-VPS and MINLP strategies, respectively.

When encountering level-*2* obstacle platoons, the FIFO-VPS strategy determined vehicle passage order based on the current virtual position projection. Once an obstacle vehicle attempted to preempt passage, FIFO-VPS yielded the right-of-way. This rule-based approach ensured operational safety but resulted in excessively large inter-vehicle spacing, causing unnecessary speed reduction in subsequent traffic flow. In contrast, the MINLP strategy exhibited a collision event, demonstrating inferior safety relative to the other two approaches. As Fig. 14(d) shows, the maximum $R_{TTCE}(t) = 0.98$ signaled a severe collision risk. This outcome arose because level-*2* obstacle vehicles expected the subject platoon to yield, while the nonlinear programming solver, relying solely on current vehicle state information, failed to resolve this conflict in time. Unlike the virtual platoon projection method, MINLP depended on spatial relationships to maintain safe distance constraints. However, in complex roundabout geometries, the numerous spatial constraints could not be satisfied simultaneously, leading to delayed solver responses to the proactive behaviors of level-*2* obstacle vehicles. The level-*k* game-theoretic approach did not cause significant travel time delays at the roundabout. The ego platoon tended to yield in the game-theoretic interaction, prioritizing the passage of vehicles already within the roundabout. This behavioral pattern aligned with the expected strategy derived from the game-theoretic framework. The level-*k* strategy enhanced merging efficiency by 7.6% and 4.6% compared to the other two strategies, while maintaining operational safety.

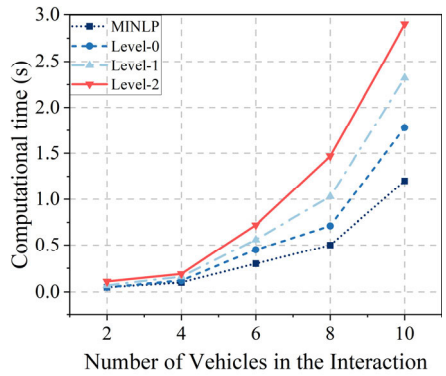

Fig. 15. Computation time of various vehicle number in the interaction.

The computation time for game-playing with different numbers of vehicles was tested. The tested quantities included 2, 4, 6, 8 and 10 vehicles, with comparisons made to the MINLP method. As a result, as shown in Fig. 15, when the number of vehicles was 6 and the game interaction was at level-*2*, the computation time was 1.4s. When the number of vehicles was 10 and the game interaction was at level-*2*, the computation time was 2.9 s.

As the number of game participants and reasoning levels increases, computation time also rises. Compared to MINLP, the most time-consuming parts are the receding horizon trajectory optimization in the curvature coordinate system and the level-*k* game iterations. the distributed computing method allows each vehicle to calculate game rewards with neighboring vehicles simultaneously. This reduces the computational load on each onboard unit compared to centralized methods, making it theoretically possible to extend the level-*k* game to *n* vehicles. However, considering planning-task implementation and communication delays, it is recommended that the number of game-participating vehicles $n \leq 6$. On one hand, in areas such as roundabouts with high road curvature, an excessively long platoon can cause significant forward-shifting of subsequent vehicles' projected coordinates on the circular road, leading to invalid projections. On the other hand, engaging in the game too early doesn't notably improve the payoff. The movement order of participating vehicles depends on the vehicles ahead and behind in their current lane.

E. Discussions

This study introduces a game-theoretic method for CAV platoon merging at unsignalized roundabouts. Its core capability lies in estimating the reasoning levels of surrounding vehicles and adaptively adjusting strategies, thereby resolving deadlocks and conflicts in mixed traffic. The formation of a stable platoon after merging prevents secondary conflicts and enhances overall safety and flow. Moreover, by incorporating an artificial potential field, the method enables personalized decision-making, allowing CAVs to dynamically tailor their behavior to diverse driving styles and risk preferences, thus ensuring robust performance in heterogeneous conditions. These features suggest significant potential for real-world

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deployment in urban traffic networks, facilitating safer and more efficient traffic management.

Nevertheless, several limitations remain. First, the hyperparameters of the potential field function are determined primarily based on references and simulation experience, and it remains uncertain whether these hyperparameters accurately capture human drivers' perception of potential driving risks. Future work will incorporate naturalistic driving data and inverse reinforcement learning (IRL) to establish a more human-like, data-driven paradigm for risk perception. Second, the current game-theoretic framework does not consider lane-changing maneuvers. In real merging scenarios, human drivers may yield to entering vehicles by changing lanes, which could lead to additional conflicts with vehicles in the inner lane. Such maneuvers pose greater challenges to computational efficiency and the attainability of Nash equilibria. Therefore, future research should integrate lateral maneuvers (e.g., lane changes and lane shifts) into the game-theoretic model and explore distributed game-solving techniques and efficient approximation algorithms to ensure real-time applicability and stability under complex traffic conditions.

V. Conclusion

In the current research, an innovative approach for the rapid merging of vehicles at roundabout intersections that synergistically combines level-k game-theoretic reasoning with the principle of social optimality is proposed. By utilizing a transformation within the Frenet coordinate system, vehicles on lanes of diverse curvatures are projected onto a singular reference trajectory, culminating in the formation of a virtual platoon. An enhanced artificial potential field function is integrated to act as the payoff function for the game, facilitating the distribution of roles—designating leaders and followers—within the virtual platoon. The real-time performance and effectiveness of the proposed strategy were validated on a hardware-in-the-loop simulation platform with a multi-platoon merging scenario at a roundabout. The study has led to the following conclusions:

1. **Adaptive Control Strategy:** The control strategy presented in this manuscript demonstrates the ability to adaptively address intelligent agents across a spectrum of reasoning levels and driving styles, facilitating rapid merging at roundabout intersections within a timeframe of 6 s. The resulting vehicle platoon maintains a consistent intervehicle distance throughout the subsequent traversal of the roundabout, with the maximum tracking error remaining below 0.2 m.

2. **Scalability of Game-Theoretic Method:** The game-theoretic method introduced in this study is scalable, enabling the intricate multiagent interaction scenario to be broken down into several fundamental three-agent interaction scenarios for queue sequencing. The ego vehicle employs Bayesian probability updates to dynamically adjust its reasoning level, opting for an automatic downgrade when surrounding agents are identified to possess superior reasoning capabilities. This adaptive strategy reduces the duration of computational iterations.

3. **Comparison with the FIFO-VPS and MINLP Strategies:** Compared to the FIFO-VPS and MINLP strategies, the control strategy proposed in this manuscript significantly reduces the incidence of deadlocks and collision at roundabout intersections while concurrently enhancing the efficiency of traffic circulation within the roundabout. In terms of traversal time, when faced with intelligent agents of varying reasoning levels, the level-k game strategies produced a 3.2% and 2.6% increase against the level-*0* obstacle vehicle platoon, 21.1% and 10.7% against the level-*1* obstacle vehicle platoon, and 7.6% and 4.6% against the level-*2* obstacle vehicle platoon.

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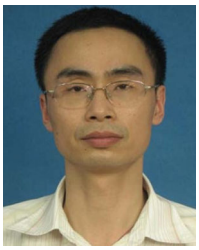

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